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Non-Human Identity Governance Agent

MAJOR PROJECT REPORT
(CS481P)

Submitted by,

Umang Mishra **1RV22CS220**

Under the guidance of

Dr. Chethana Murthy
Associate Professor
Dept. of CSE
RV College of Engineering

Mr. Tejeshwar Rao
Manager DevSecOps & Cloud
DevSecOps & Cloud Engineering
Honeywell Technology Solutions

In partial fulfillment for the award of degree of

Bachelor of Engineering
in
Computer Science and Engineering

Department of CSE
2025-26

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Go, change the world

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(*Autonomous institution affiliated to VTU, Belagavi*)

Department of Computer Science and Engineering

CERTIFICATE

Certified that the major project (CS481P) work titled ***Non-Human Identity Governance Agent*** is carried out by **Umang Mishra** (**1RV22CS220**) who is bonafide student of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in **Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

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DECLARATION

I, **Umang Mishra** student of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Non-Human Identity Governance Agent**' has been carried out by myself and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Computer Science and Engineering** during the year 2025-26.

Further I declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

I also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and I will be one of the authors of the same.

Place: Bengaluru

Date:

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ABSTRACT

With cloud-native architectures and DevOps practices becoming the norm, machine credentials (service principals, API keys, GitHub tokens, managed identities) have grown rapidly and now outnumber human users by as much as 144:1 in many enterprises. Yet these Non-Human Identity (NHI) tend to get far less governance attention than human accounts, leaving a blind spot that attackers routinely exploit through credential-based attacks. According to the OWASP NHI Top 10 (2025), roughly 70% of organisations still have plaintext secrets sitting in their source repositories.

To tackle this problem, the project built an AI-powered NHI Governance Agent that discovers, scores, governs, and rotates non-human identities across Azure and GitHub. The design follows a four-layer modular approach: a discovery layer that pulls data through the Microsoft Graph API and OpenSSL TLS probes, a classification layer driven by LangChain agentic orchestration, a policy enforcement layer built on Open Policy Agent with Rego rules, and a remediation layer that triggers credential rotation via Azure Key Vault and GitHub Actions workflows.

In its current deployment the system monitors TLS certificate expiry across production endpoints and checks Azure AD service principal credentials on a schedule using GitHub Actions. It generates timestamped HTML reports with colour-coded status indicators, applies a three-tier classification (CRITICAL, EXPIRING, OK) to help operators focus on what matters most, and sends SMTP email alerts only when credentials actually enter the risk window. The monitoring modules reached 82% unit test coverage during validation.

By removing long-lived credentials and enforcing continuous least-privilege policies, the agent shrinks the blast radius of credential-based attacks in a measurable way. It also produces audit-ready NHI inventory reports that map directly to SOC 2 and ISO 27001 compliance requirements.

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ABBREVIATIONS

Abbreviation	Full Form
AD	Azure Active Directory
API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
CISO	Chief Information Security Officer
CSA	Cloud Security Alliance
CSV	Comma-Separated Values
CVE	Common Vulnerabilities and Exposures
GA	GitHub Actions
HTML	HyperText Markup Language
IAM	Identity and Access Management
JSON	JavaScript Object Notation
ML	Machine Learning
NHI	Non-Human Identity
OPA	Open Policy Agent
OWASP	Open Worldwide Application Security Project
PAT	Personal Access Token
REST	Representational State Transfer
SDK	Software Development Kit
SIEM	Security Information and Event Management
SMTP	Simple Mail Transfer Protocol
SOC	System and Organisation Controls
SP	Service Principal
TLS	Transport Layer Security
YAML	YAML Ain't Markup Language
ZTA	Zero Trust Architecture

Chapter 1

Introduction

CHAPTER 1

INTRODUCTION

This chapter explains why unmanaged Non-Human Identity (NHI) in today's cloud environments pose a real and growing threat. It lays out the motivation behind building an AI-powered governance agent, defines what the project aims to achieve, and gives a quick overview of how the rest of the report is structured.

1.1 Overview

Over the past few years, organisations have moved aggressively towards cloud-native architectures, DevOps pipelines, and microservice-based systems. A natural consequence of this shift is that machine-to-machine communication has exploded in volume, and with it the number of NHIs, including service principals, GitHub tokens, Application Programming Interface (API) keys, and managed identities. In many large enterprises these machine credentials now outnumber human user accounts by anywhere from 10:1 to 144:1, yet they rarely get the same level of scrutiny or governance [1]. This mismatch creates a dangerous blind spot, because long-lived, unmanaged, or over-privileged machine credentials are exactly what attackers look for.

The fallout from this governance gap can be severe. Credential-based attacks are consistently among the top breach vectors worldwide; the Verizon 2023 Data Breach Investigations Report found that credential abuse was the primary path into the victim's environment in 49% of all recorded breaches. A single leaked Service Principal (SP) with broad Azure Active Directory (AD) permissions is enough to let an attacker compromise an entire tenant, move laterally across subscriptions, and steal sensitive data at cloud scale.

The IA-DevSecOps-Identity-Governance project was built to address this head-on. It delivers an AI-powered NHI Governance Agent that continuously discovers, classifies, governs, and rotates non-human identities across Azure and GitHub. Unlike static secrets scanners or one-off audit tools that wait for a human to act, this agent runs autonomously, relying on policy-as-code enforcement, anomaly detection, and automated credential rotation to keep things secure without constant manual oversight.

1.2 Problem Statement

The way most organisations handle NHI governance today suffers from three core problems.

The first is that monitoring is reactive and point-in-time. Most tools only catch expiry risks or exposed secrets after the damage is done, and there is no automated way to fix things. Operators have to manually respond to alerts, which introduces delays and leaves the organisation exposed between audit cycles.

The second issue is that the tooling is completely fragmented. Certificate monitors work independently of Azure credential tools, and neither has any visibility into GitHub tokens or managed identities. There is no single system that maintains a unified, cross-platform view of all non-human identities, including the dormant or orphaned ones that tend to get forgotten.

Third, current solutions lack any real risk intelligence. They rely on simple binary checks (expired or not expired) that miss a lot of nuance. For example, a service principal that has not been used in ninety days but is still valid often poses a higher risk than one that is actively used but expiring in forty-five days. Yet existing tools treat both situations the same way.

The Open Worldwide Application Security Project (OWASP) Non-Human Identities Top 10 report from 2025 paints a stark picture: 70% of organisations have plaintext secrets sitting in their code repositories and 47% have unused Identity and Access Management (IAM) roles that are still active [2]. Similarly, a Cloud Security Alliance (CSA) survey from 2024 found that only 15% of organisations feel confident they can prevent NHI attacks [3].

1.3 Objectives

The project is built around five key objectives. First, it aims to continuously discover and maintain a live inventory of every NHI across Azure and GitHub, including dormant or orphaned identities that slip through conventional audit processes [3]. Second, it implements an AI-driven risk scoring engine using LangChain that evaluates each identity based on sensitivity, usage patterns, and privilege scope, drawing on proven anomaly detection techniques [4], [5]. Third, it enforces policy-as-code rulesets through Open Policy Agent (OPA) with Rego, so that governance requirements are codified in a declarative, version-controlled format rather than buried in documentation [6]. Fourth, it triggers

automated credential rotation through GitHub Actions (GA) whenever a policy violation is detected, without waiting for someone to manually intervene. This brings the window of credential exposure down from months to hours [7]. Fifth and finally, it generates audit-ready NHI inventory reports and compliance dashboards that support SOC 2 and ISO 27001 frameworks [8], [9].

1.4 Scope

The project was executed in two phases.

In the **current implemented phase**, the system monitors Transport Layer Security (TLS) certificate expiry for production-grade domain endpoints and checks Azure AD service principal credentials on a regular schedule using GitHub Actions workflows. It produces timestamped HTML reports with rolling retention, uploads them as 90-day GitHub Actions artifacts, and fires conditional Simple Mail Transfer Protocol (SMTP) email alerts whenever credentials enter the expiry risk window.

The **proposed evolution phase** takes things further by adding autonomous NHI discovery through the Microsoft Graph API and GitHub Representational State Transfer (REST) API, multi-dimensional risk scoring driven by LangChain agentic orchestration, policy-as-code enforcement using OPA and Rego, automated credential rotation via Azure Key Vault, and a real-time Azure-hosted NHI inventory dashboard.

1.5 Societal Relevance and SDG Alignment

The project connects to two United Nations Sustainable Development Goals. It supports **SDG 9** (Industry, Innovation, and Infrastructure) by creating resilient, secure, and automated governance infrastructure for the digital world. It also contributes to **SDG 16** (Peace, Justice, and Strong Institutions) by promoting institutional accountability in how machine identities are managed and producing audit trails that satisfy regulatory compliance needs.

1.6 Report Organisation

The rest of this report is laid out as follows. Chapter 2 covers the literature survey, looking at the NHI threat problem, AI-driven anomaly detection, policy-as-code frameworks, and the existing tools in the space. Chapter 3 goes into the system design, covering the four-layer architecture, component descriptions, methodology, and security design principles. Chapter 4 walks through the implementation, including the development

environment, runtime execution flow, and how each system layer was built. Chapter 5 presents the results and discussion, covering unit test coverage, performance benchmarks, policy engine validation, and a comparison with other tools. Chapter 6 wraps up with conclusions, future scope, and what was learned along the way.

In short, this chapter laid out why an AI-powered NHI Governance Agent is needed: machine identities are growing fast, credential-based attacks are a serious threat, and the tools currently available are simply not up to the task. Five clear objectives were defined, spanning discovery, risk scoring, policy enforcement, automated rotation, and compliance reporting. The next chapter surveys the academic literature, industry reports, and existing tools that shaped the design of this system.

Chapter 2

Literature Survey

CHAPTER 2

LITERATURE SURVEY

This chapter takes a close look at what has already been written, built, and reported on when it comes to governing non-human identities, using AI for security automation, and enforcing policies through code. The goal here is to understand the groundwork that exists and, just as importantly, to spot the gaps that the proposed NHI Governance Agent is meant to fill.

2.1 Introduction to the Survey

The reading for this review fell into three broad areas. First, there is the academic side: papers on identity governance, anomaly detection, and zero-trust architectures. Then there are industry reports and surveys that put hard numbers on how big the NHI problem really is and how poorly most organisations are handling it. Finally, there are the commercial tools that already tackle parts of this problem, each with its own strengths and blind spots. Rather than going through every source one by one, this chapter pulls findings together by theme so that a clearer overall picture emerges of where things stand and where there is room to do better.

2.2 NHI Threat Problem and Scale

One of the first things that becomes obvious when researching this area is just how many non-human identities modern organisations are dealing with. Cloud-native architectures, microservice deployments, and DevOps automation have all driven an enormous increase in machine credentials. According to enterprise-wide analyses, NHIs outnumber human identities by ratios anywhere from 10:1 up to 144:1, and that number is growing at roughly 44% year over year [1]. The reason is straightforward: every service principal, API key, deploy token, managed identity, and Continuous Integration / Continuous Deployment (CI/CD) pipeline secret adds another non-human identity to the mix, and modern cloud setups rely on huge numbers of them for machine-to-machine communication.

From a security standpoint, this growth is alarming. Credential-based attacks consistently show up as one of the top breach vectors worldwide; in fact, credential abuse has been identified as the primary attack path in close to half of all recorded data breaches [10]. The OWASP Non-Human Identities Top 10 report lays out the most critical risks

organisations face, including things like failing to properly offboard machine credentials, secrets leaking into source code repositories, and service accounts that have far more permissions than they need [2]. That same report found some striking numbers: 70% of organisations have plaintext secrets sitting in their code repositories, and 47% still have unused IAM roles that remain active and could be exploited.

Practitioner surveys paint an equally concerning picture. One survey covering more than 800 IT and security professionals found that just 15% of organisations feel confident they can prevent NHI-related attacks, while a full 69% express real worry about their machine identity security posture [3]. The biggest pain points they flag are poor auditing capabilities, weak privilege management, and a lack of automated policy enforcement for non-human credentials. Analyst reports on the NHI management market back this up, noting that the machine-to-human identity ratio tops 17:1 even in mid-sized organisations, and projecting strong market growth as enterprises start treating NHI governance as a board-level security priority [11], [12]. Established threat intelligence frameworks also confirm that attackers actively go after long-lived, unrotated machine credentials because they offer a reliable way to maintain persistent access in compromised environments [13].

2.3 Zero Trust and Identity Access Management Frameworks

Zero trust has become the go-to security model for modern cloud environments, and for good reason. Instead of trusting anything inside a network perimeter, it works on the assumption that every access request needs to be verified, regardless of where it comes from. The foundational architecture sets out three core ideas: verify explicitly, enforce least privilege, and always assume breach [14]. These principles were originally aimed at human users, but they arguably matter even more for non-human identities. After all, machine credentials operate autonomously, often carry elevated privileges, and do not have the kind of behavioural oversight that human users get.

When it comes to identity and access management in the cloud, researchers have looked at the challenges that come with dynamic resource provisioning and short-lived workloads. Work examining IAM in cloud settings through a Zero Trust Architecture (ZTA) lens reveals a real tension: on one hand, automated credential provisioning is essential for agile development; on the other, continuous verification and least-privilege

enforcement cannot be compromised [15]. Some researchers have proposed cross-cloud federated IAM systems that work across multiple cloud providers by computing trust scores and modelling behaviour to enforce least privilege across cloud boundaries [16]. Broader surveys of IAM in multi-cloud environments make it clear that current identity governance practices are fragmented, and keeping consistent security policies across different cloud platforms remains a real struggle [17].

The rise of autonomous AI agents adds yet another layer of complexity. Research on zero-trust identity frameworks for agentic AI shows that traditional authentication protocols like OAuth and OpenID Connect were not designed for agents that make their own decisions and interact with external systems independently [18]. These studies propose decentralised approaches using verifiable credentials and context-aware access control, which lay useful theoretical groundwork for governing the growing number of AI-driven NHIs in enterprise settings. Along similar lines, work on decentralised identity management brings together self-sovereign identity principles, decentralised identifiers, and zero-knowledge proofs in multi-domain orchestration frameworks, offering a cryptographically verifiable foundation for managing NHI lifecycles across distributed systems [19].

The standards and guidelines that shape this space also matter. Digital identity guidelines from standards bodies set out assurance levels and identity proofing requirements that, while mainly written with people in mind, still inform how strong the authentication and credential management should be for machine identities [20]. International information security management standards provide the compliance frameworks that any NHI governance solution ultimately has to measure up against, especially around access control, cryptographic key management, and operational security [8], [9].

2.4 AI-Driven Trust Assessment and Anomaly Detection

Machine learning has increasingly found its way into security operations, mainly because static, rule-based monitoring just cannot keep pace with sophisticated and constantly evolving threats. Broad surveys of anomaly detection methods break these approaches down into statistical, classification-based, clustering-based, and information-theoretic categories, providing a solid theoretical basis for spotting unusual behaviour in complex systems [21]. One algorithm that has proven especially useful is the isola-

tion forest, which works by measuring how many random partitions it takes to isolate a data point. It turns out to be very effective at picking out outlier behaviour in high-dimensional datasets, which is exactly the kind of data that NHI access patterns generate [4].

There is also interesting research on dynamic trust assessment powered by AI. Some proposed frameworks use a mix of supervised and unsupervised Machine Learning (ML) models to generate trust scores that update continuously, rather than relying on a simple yes-or-no authentication check [5]. These trust values shift based on behavioural signals, environmental context, and past access patterns, which makes them much more nuanced. For NHI governance, this kind of approach is valuable because it can distinguish between, say, a service principal that has been sitting idle for ninety days with elevated privileges and one that is actively used but nearing credential expiry. Those are very different risk profiles, and a static rule would treat them the same way.

The wider body of work on ML in security operations has been covered in several surveys, and they are honest about both the potential and the pitfalls [22]. Labelled training data for rare security events is hard to come by. False positives in a live operational setting are expensive, both in analyst time and in alert fatigue. And there is a real need for models that can explain why they flagged something, not just that they did. Looking ahead, the emergence of large language model-based autonomous agents opens up new possibilities for security orchestration. These agentic AI systems can reason through complex security scenarios, pull together information from multiple sources, and carry out multi-step remediation workflows [23].

2.5 Policy-as-Code and Governance Automation

Policy-as-code is one of those ideas that sounds simple but represents a real shift in how governance works. Instead of writing rules in documents that someone has to read and manually enforce, policies get expressed as executable code that can be version-controlled, tested, and enforced automatically. Reviews of policy-as-code approaches for cloud-native security consistently point to OPA and its Rego policy language as the leading framework for this kind of thing [6], [24]. With OPA, an organisation can write rules like "credentials must not exceed ninety days of age" or "service accounts must follow least-privilege scoping," and those rules run automatically as part of CI/CD pipelines. That is a major step up from hoping someone remembers to check a spreadsheet.

Managing secrets and credentials at scale is another area that has received a lot of attention in the literature. Systematic reviews on software secret management describe what researchers call secret sprawl, where credentials end up scattered across code repositories, configuration files, and even chat tools, with no central control [25]. Research on automated credential lifecycle management in enterprise cloud environments shows that when rotation is handled programmatically, integrated with cloud provider APIs and vault services, the exposure window for a compromised credential can shrink from months down to hours [7]. Standards bodies have also published detailed recommendations for cryptographic key management, covering rotation schedules, key lengths, and lifecycle practices, all of which feed directly into the policy rules an NHI governance system would enforce [26].

On the DevOps side, the integration of security into delivery pipelines (often called DevSecOps) has been studied through large-scale practitioner surveys. Reports on DevSecOps adoption show that organisations embedding automated security checks and policy enforcement into their pipelines see noticeably faster remediation times and fewer vulnerabilities [27]. Performance research confirms this as well: the highest-performing technology organisations, the ones with frequent deployments and low failure rates, tend to invest heavily in automated governance controls that cut down on manual work while making security stronger [28]. And in microservice architectures specifically, where the number of inter-service communication channels and their associated credentials grows quadratically with the number of services, these challenges become even more pressing [29].

2.6 Existing Tools and Their Limitations

There are already a number of commercial and open-source tools that deal with parts of the NHI governance problem. However, after reviewing them, it becomes clear that none of them offers the kind of comprehensive, autonomous governance capability that modern enterprises actually need.

HashiCorp Vault is probably the most well-known secrets management platform out there. It handles dynamic secret generation, provides encryption as a service, and supports fine-grained access policies tied to identity [30]. As a centralised secrets store, it is excellent, and it does support programmatic credential rotation through its API. That said, Vault was not built to autonomously discover NHIs scattered across multi-platform

environments. It does not do AI-driven risk scoring of credential postures, and it certainly does not act as an independent governance agent that can find and fix policy violations on its own without someone stepping in.

Microsoft Entra Workload ID takes a different angle, providing identity services specifically for workloads running in Azure. It supports managed identities and federated credentials for cloud-native applications and offers basic lifecycle features like credential expiry notifications for Azure-native service principals [31]. The catch is that its scope stops at the Azure ecosystem. It does not correlate NHIs across GitHub tokens, CI/CD pipeline secrets, or credentials from third-party services. And it does not bring any AI-driven risk classification or anomaly detection to the table.

What both of these tools, along with the other solutions reviewed, have in common is that they each handle one or two pieces of the NHI governance lifecycle, whether that is discovery, storage, rotation, or monitoring, but none of them ties all four together into a single continuously operating autonomous agent. Research into certificate lifecycle management challenges reinforces this point, showing that when expiry monitoring, policy enforcement, and automated renewal are spread across disconnected tools, operational gaps inevitably appear [32].

2.7 Research Gap and Rationale

After going through the literature, one thing stands out clearly: there is a well-documented gap in how NHI security is handled today. On the academic side, researchers have laid strong theoretical groundwork for zero-trust identity frameworks, AI-driven anomaly detection, and policy-as-code enforcement. On the industry side, reports have put real numbers to the problem, showing that unmanaged machine credentials are one of the most significant attack surfaces in enterprise environments right now. But when it comes to actual solutions, whether from academia or the commercial market, nothing brings everything together. No existing tool or prototype combines autonomous cross-platform NHI discovery, AI-driven continuous risk scoring, policy-as-code enforcement with machine-executable rules, and automated credential rotation in a single, unified governance agent.

That is precisely the gap this project aims to close. The proposed NHI Governance Agent brings together LangChain-based agentic AI orchestration for intelligent risk assessment, OPA Rego policies for codified governance enforcement, and GA workflows for

automated remediation, capabilities that have until now only existed separately. The agent is designed to run continuously and autonomously, removing the manual bottlenecks and the overhead of coordinating between multiple disconnected tools that define current approaches.

2.8 Feasibility Study

From a technical standpoint, this project is very much feasible because all the key building blocks are readily available. The Azure Software Development Kit (SDK) for Python and the Microsoft Graph API give full programmatic access to Azure Active Directory service principals, application registrations, and managed identities [33], [34]. On the GitHub side, the REST API makes it possible to enumerate deploy keys, Actions secrets, and repository-level NHIs [35]. For the AI-driven risk scoring and decision-making side of things, LangChain provides the agentic orchestration framework needed [36], while OPA with Rego handles the declarative policy-as-code enforcement [24]. Azure Key Vault takes care of secure credential storage and rotation [37], and GA provides the CI/CD automation platform to run governance workflows on whatever schedule is needed [38]. Python 3.11 was chosen as the primary language because of its mature async support and the extensive library ecosystem available for all the integrations this project requires [39].

To sum up, this chapter reviewed the academic research, industry reports, and commercial tools that are relevant to NHI governance. The key takeaways are that the NHI threat problem is both severe and growing fast, that zero-trust and AI-driven anomaly detection provide the theoretical basis for smarter identity governance, and that policy-as-code enables automated enforcement of governance rules. Most importantly, the review confirmed a clear gap: nothing out there currently integrates cross-platform discovery, AI-driven risk scoring, policy enforcement, and automated rotation into a single autonomous agent. The feasibility study showed that all the required technologies are mature and accessible. The next chapter presents the system architecture designed to address this gap.

Chapter 3

System Design

CHAPTER 3

SYSTEM DESIGN

This chapter walks through how the NHI Governance Platform was designed from the ground up. It covers the four-layer architecture that organises the system, explains the individual components and how they fit together, lays out the methodology used to plan the work, and discusses both the system requirements and the security principles that shaped every design decision.

3.1 Architectural Overview

The NHI Governance Platform uses a four-layer modular architecture. Each layer handles a distinct concern (configuration, monitoring and intelligence, output and reporting, or orchestration) so they stay cleanly separated. The practical benefit of this is that any single layer can be updated, tested, or swapped out on its own without breaking anything else in the system.

Figure 3.1 shows how these layers are stacked.

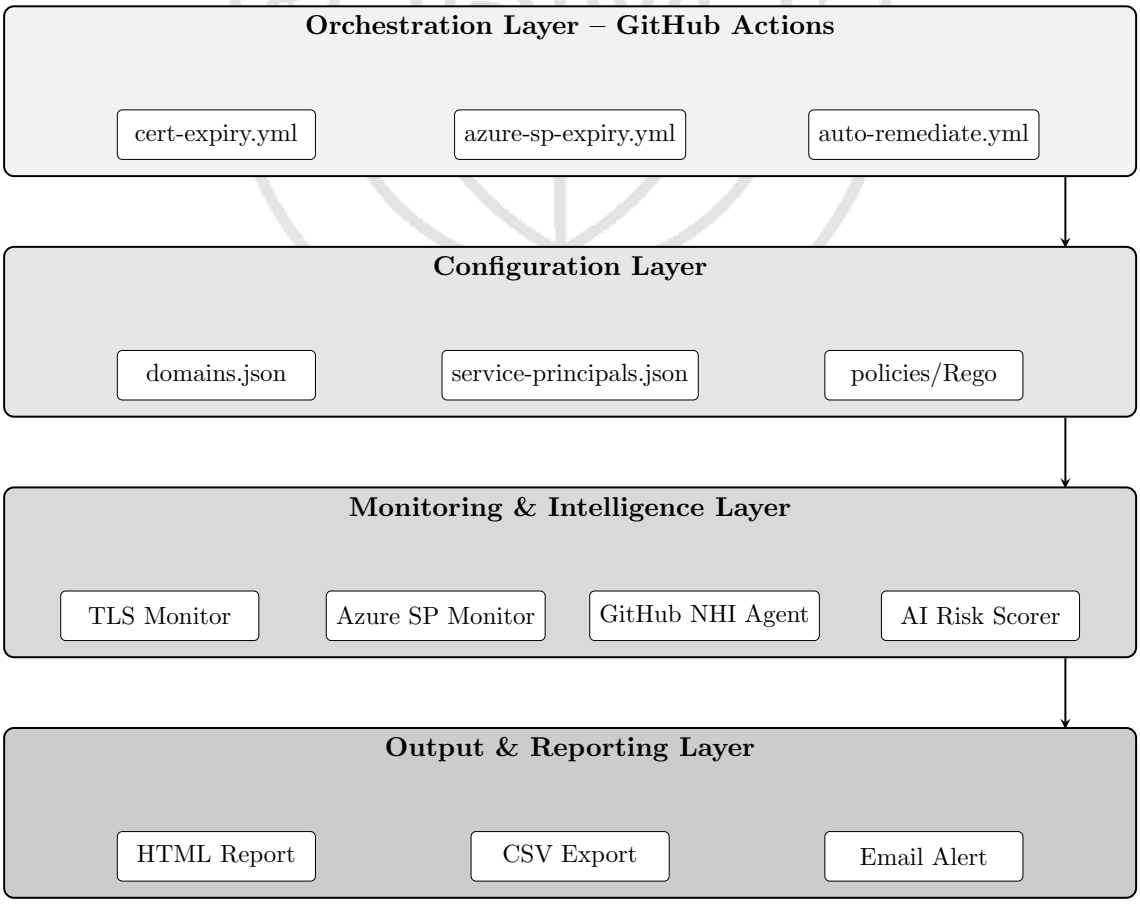

Figure 3.1: Four-Layer System Architecture of the NHI Governance Platform

One thing that was important during design was making sure the architecture could grow. Right now it handles the operational deployment, but the AI-powered features planned for the future can be plugged in as extra layers without having to rework what already exists.

Figure 3.2 breaks down the full governance workflow step by step, tracing how data moves from initial discovery all the way through classification, policy checks, and finally automated remediation.

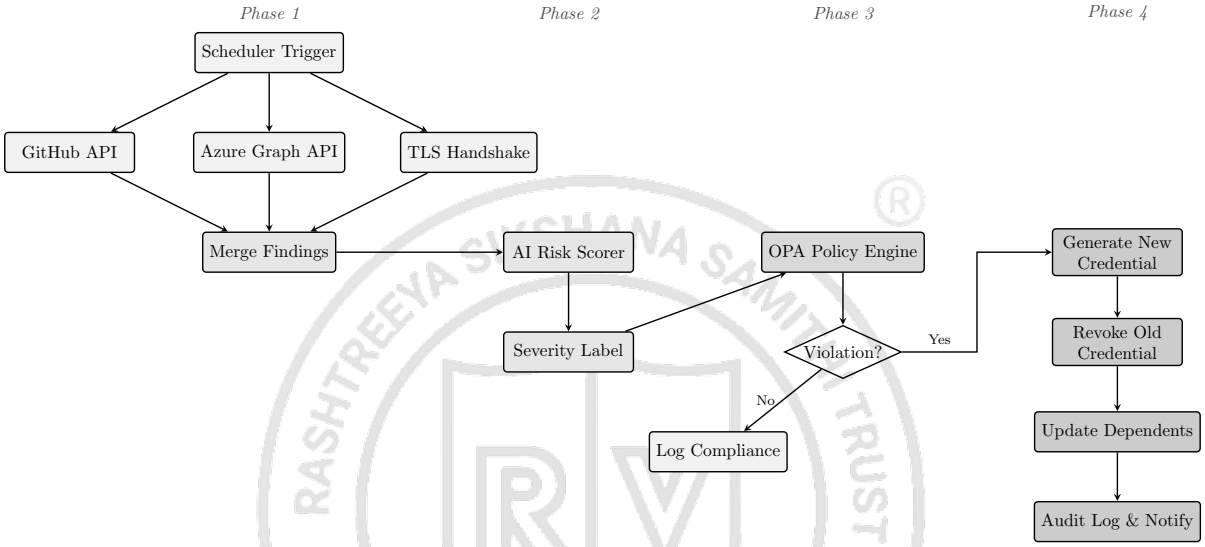

Figure 3.2: End-to-End Governance Workflow: Discovery to Remediation

3.2 Methodology and Planning of Work

The project was planned around three distinct phases, each building on the previous one. The goal was to keep security and compliance running continuously across the entire NHI lifecycle rather than relying on one-off audits.

Phase 1: Continuous Discovery and Inventory. In this first phase, the system connects to Azure Active Directory (Entra ID), Azure Key Vault, and the GitHub API to pull together a live inventory of every NHI in the environment. It keeps scanning these ecosystems on an ongoing basis and picks up metadata like credential type, when it was created, when it was last used, what level of access it has, and who owns it.

Phase 2: Classification and Anomaly Detection. Once the inventory is built, the system needs to make sense of it. LangChain handles the agentic orchestration here, analysing how each identity behaves based on the collected metadata. An AI-driven risk scoring engine then classifies identities according to how sensitive they are, what

privileges they hold, and their usage history. If something looks unusual, the system flags it straight away.

Phase 3: Automated Governance and Remediation. The last phase is where enforcement happens. Governance rules are written as infrastructure-as-code policies using OPA and Rego, which turns organisational requirements into rules a machine can execute. When a policy threshold gets breached, say a credential is older than 30 days or an identity has sat unused for more than 7 days, GA pipelines kick in automatically to rotate or remediate.

3.3 System Requirements Specification

3.3.1 Hardware Requirements

Since everything runs in the cloud, there is no on-premise hardware to worry about. The system is hosted entirely on Microsoft Azure. Compute comes from Azure virtual machines or containers that run the Python-based workflows, and storage for things like inventory snapshots, policy logs, and audit records is handled through Azure Monitor and Azure's cloud storage services.

3.3.2 Software Requirements

Table 3.1 lists the main technologies the system depends on.

Table 3.1: Software Requirements

Component	**Technology**
Programming Language	Python 3.11, Bash
APIs and SDKs	Azure SDK, Microsoft Graph API, GitHub REST API
Cloud Services	Azure Key Vault, Azure Managed Identities, Azure Monitor
AI and Orchestration	LangChain, Open Policy Agent (Rego)
CI/CD Automation	GitHub Actions

3.3.3 Functional Requirements

The system needs to continuously find every NHI across Azure AD and GitHub, including dormant and orphaned identities that would be missed by a simple point-in-time audit [2]. Once discovered, each identity gets classified using a three-tier threshold model (CRITICAL, EXPIRING, and OK) based on how close the credential is to expiring. On top of that, the system calculates an AI risk score for each identity by looking at privilege scope, how old the credential is, and usage patterns, then rolling those factors into a single

normalised metric [5]. When a policy violation is detected, the system should be able to rotate credentials automatically, without any human pressing a button, using Azure Key Vault for secure credential generation [37]. All results get presented through HTML and CSV reports with colour-coded statuses so that both a person reviewing them and a Security Information and Event Management (SIEM) tool can make sense of the data.

3.3.4 Non-Functional Requirements

Reliability was a big consideration. The system should run continuously with no scheduled downtime so that governance cycles fire on schedule every single time. For security, all state management follows least-privilege access controls, meaning the monitoring identity only gets the bare minimum permissions it needs to perform discovery [14]. Scalability also matters because an enterprise environment could easily have thousands of identities spread across hundreds of repositories. The system has to handle that kind of volume without slowing down, and it supports this through parallel agent execution and incremental syncing with delta queries.

3.4 Component Design

Figure 3.3 shows how the modules depend on each other and how data flows between them.

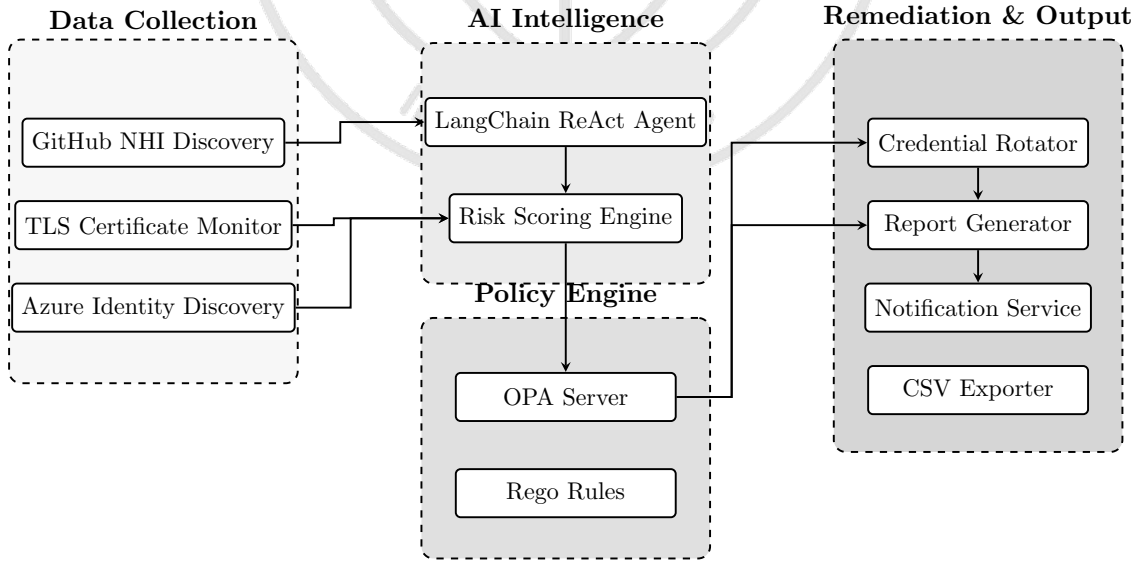

Figure 3.3: Component Dependency and Data Flow Diagram

3.4.1 Configuration Layer

A key design choice was to pull all monitoring targets out of the code and into versioned JSON files. This way, an operator can add or remove targets without ever touching the core scripts. For instance, `data/domains.json` holds the TLS endpoints and supports things like custom port numbers and SNI override objects. Similarly, `data/service-principals.json` lists the Azure AD applications and uses dual-indexed filtering to keep lookups flexible.

3.4.2 Monitoring and Intelligence Engine

The **TLS Certificate Monitor** works by opening a live TLS handshake using OpenSSL's `s_client` command to grab the certificate expiry information. If a particular domain fails, the monitor catches that error on its own and moves on. It does not kill the rest of the scan.

The **Azure Service Principal Credential Monitor** talks to the Microsoft Graph API to pull both client secret and certificate-based credentials. Any credentials that have already expired get filtered out so they do not clutter the results.

The **Threshold Classification Engine** uses a straightforward three-tier model to label credentials: anything with $\leq$ 30 days left is marked CRITICAL, $\leq$ 60 days is EXPIRING, and everything beyond 60 days is OK.

3.4.3 Output and Reporting Layer

Reports come out as self-contained HTML files with timestamps and colour-coded statuses so it is easy to spot problems at a glance. The system keeps only the fifteen most recent reports and deletes older ones automatically. CSV exports are also generated for teams that want to feed the data into a SIEM. On top of that, SMTP email notifications go out when certain conditions are met.

3.4.4 GitHub Actions Orchestration Layer

Three GA workflows drive the platform's automation [38]. The first, `cert-expiry.yml`, runs the TLS certificate monitoring on a weekly schedule. It scans every configured domain and produces an expiry report. The second, `azure-sp-expiry.yml`, checks Azure service principal credentials every day by authenticating through the Microsoft Graph API to pull credential metadata [34]. The third, `automate-build.yml`, is the full DevSecOps pipeline that fires on every push to run code quality checks, security scans, and

automated tests. All three workflows run on a private self-hosted runner pool so that execution stays inside the organisation's secure network.

3.5 Security Design Principles

Security was not an afterthought. It shaped the design from the start, following zero trust architecture principles [14]. No credentials are ever hardcoded into the source code. Instead, all secrets live as encrypted secrets at the GitHub repository level, so sensitive values never end up in version control [25]. The monitoring identity itself only has read-only access (`Application.Read.All`), sticking strictly to least privilege. Azure client secret values are never written to logs or shown in reports, which prevents credentials from leaking through output channels. Every workflow runs exclusively on the private self-hosted runner pool, meaning governance tasks execute inside the organisation's own network rather than on shared public infrastructure. Finally, generated reports are listed in `.gitignore` so that sensitive NHI inventory data cannot accidentally get committed and pushed.

3.6 Risk Scoring Algorithm Design

One of the most important parts of the system is the risk scoring engine. It uses a deterministic, multi-factor algorithm that gives each discovered NHI a score between 0 and 100. The algorithm looks at four weighted factors, and because it is deterministic, the same inputs always produce the same score, which makes results consistent and easy to audit across different credential types.

3.6.1 Scoring Components

The overall risk score is just the sum of four individual component scores:

$$S_{total} = S_{age} + S_{type} + S_{privilege} + S_{ssl} \tag{3.1}$$

where each component is calculated as follows:

Age Component (40 points maximum): Measures credential age relative to policy thresholds. The score scales linearly from 0 (new credential) to 40 (at or beyond threshold age):

$$S_{age} = \min\left(\frac{age_{days}}{threshold_{days}}, 1.0\right) \times 40 \tag{3.2}$$

Type Weight (30 points maximum): Assigns risk weight based on credential type

sensitivity:

- Personal Access Token (PAT): 30 points
- Deploy Key (write access): 30 points
- Deploy Key (read-only): 18 points
- Actions Secret: 14 points
- SSL Certificate: 10 points

Privilege Scope (20 points maximum): Evaluates access level and permission scope. For PATs, each high-privilege scope (`repo`, `admin:org`, `workflow`, `delete_repo`, `write:packages`) adds 7 points, capped at 20. For deploy keys, write access adds 20 points.

SSL Severity Bonus (up to 90 points): Additional scoring for certificate status:

- EXPIRED/ERROR: +90 points (automatic CRITICAL)
- CRITICAL status (< 30 days): +40 points
- WARNING status (30–60 days): +20 points

3.6.2 Severity Classification

The final score (capped at 100) maps to severity labels:

Table 3.2: Risk Score Severity Classification

Score Range	Severity	Action Required
≥ 70	CRITICAL	Immediate remediation
≥ 50	HIGH	Priority rotation within 24 hours
≥ 30	MEDIUM	Scheduled rotation within 7 days
< 30	LOW	Monitor only

3.6.3 Worked Example

Consider a Personal Access Token created 180 days ago with `admin:org` scope:

- Age: $\min(180/30, 1.0) \times 40 = 40$ points
- Type: PAT = 30 points

- Privilege: `admin:org` → 20 points
- SSL: N/A → 0 points
- **Total: 90 points → CRITICAL severity**

3.7 State Diagrams

To make it easier to understand how data moves through the system and what the governance agent is doing at any given moment, this section presents the state machine models. There are two: one for the lifecycle of an individual finding, and another for the system's own execution states.

3.7.1 Finding State Lifecycle

Every NHI finding passes through four distinct states during a single governance cycle. Knowing this lifecycle helps a lot when it comes to debugging issues, auditing results, or adding new features down the line.

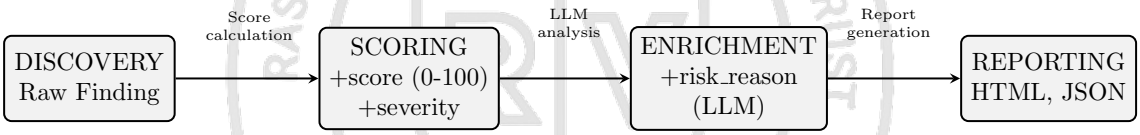

Figure 3.4: Finding State Lifecycle

State Transitions Explained:

1. **DISCOVERY → SCORING:** At this point a finding only has basic metadata: the host, credential type, creation date, and status. The scoring phase takes that raw data and runs it through the four-component risk formula (Equation 3.1) to produce a score between 0 and 100, then assigns a severity label like CRITICAL, HIGH, MEDIUM, or LOW.

2. **SCORING → ENRICHMENT:** Any finding that scores 50 or above can optionally be sent to the LLM (GPT-4o through Azure OpenAI) for a more detailed explanation. The `risk_reason` field gets filled in with a plain-English description of why the finding is risky, which is useful for reviewers who may not understand the raw scores.

3. **ENRICHMENT → REPORTING:** Once a finding is fully enriched, it gets written out in two formats: HTML for a human-readable dashboard and JSON for machine consumption. Having both means the reports work for manual reviews and can also be piped into automated SIEM workflows.

3.7.2 System Execution State Machine

The governance agent itself goes through seven execution states during a run. Breaking it down this way makes it straightforward to monitor what is happening, debug problems, and profile where time is being spent.

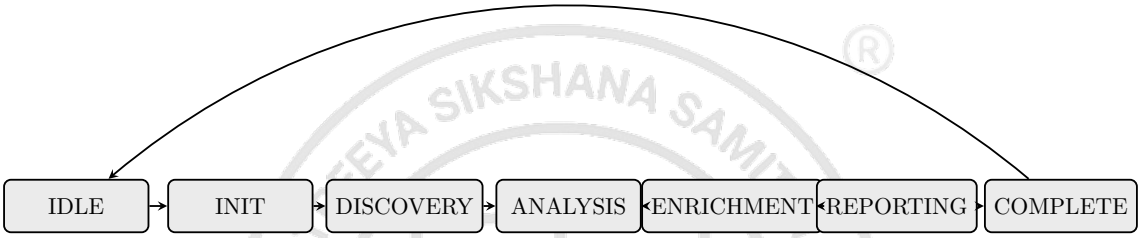

Figure 3.5: System Execution State Machine

State Descriptions:

- **IDLE:** The system is just waiting, either for someone to start a scan or for a scheduled trigger to fire. No resources are being used.
- **INIT:** Configuration gets loaded from files like `config.yaml` and `domains.json`, agents are created, and LangChain orchestration is set up ready to go.
- **DISCOVERY:** SSL certificate checks and GitHub NHI discovery run at the same time. Each has its own timeout: 10 seconds for SSL and 60 seconds for GitHub API calls.
- **ANALYSIS:** The risk scoring formula runs against all findings, and the policy engine checks rules R001 through R007.
- **ENRICHMENT:** If enabled, high-risk findings (those scoring 50 or above) get sent to GPT-4o for natural language explanations.
- **REPORTING:** HTML and JSON reports are generated, and email alerts go out for anything that triggered a policy violation.

- **COMPLETE:** The report file path is handed back and the system loops back to IDLE, ready for the next cycle.

3.8 UML Design Diagrams

This section contains the UML diagrams that formally describe the system's structure and behaviour. They act as the main design documentation and give developers, architects, and stakeholders a shared visual language to refer to.

3.8.1 Sequence Diagram: Full Orchestration

Figure 3.6 shows the complete sequence diagram for a governance scan cycle. It captures the order in which messages are exchanged and which components are active at each point.

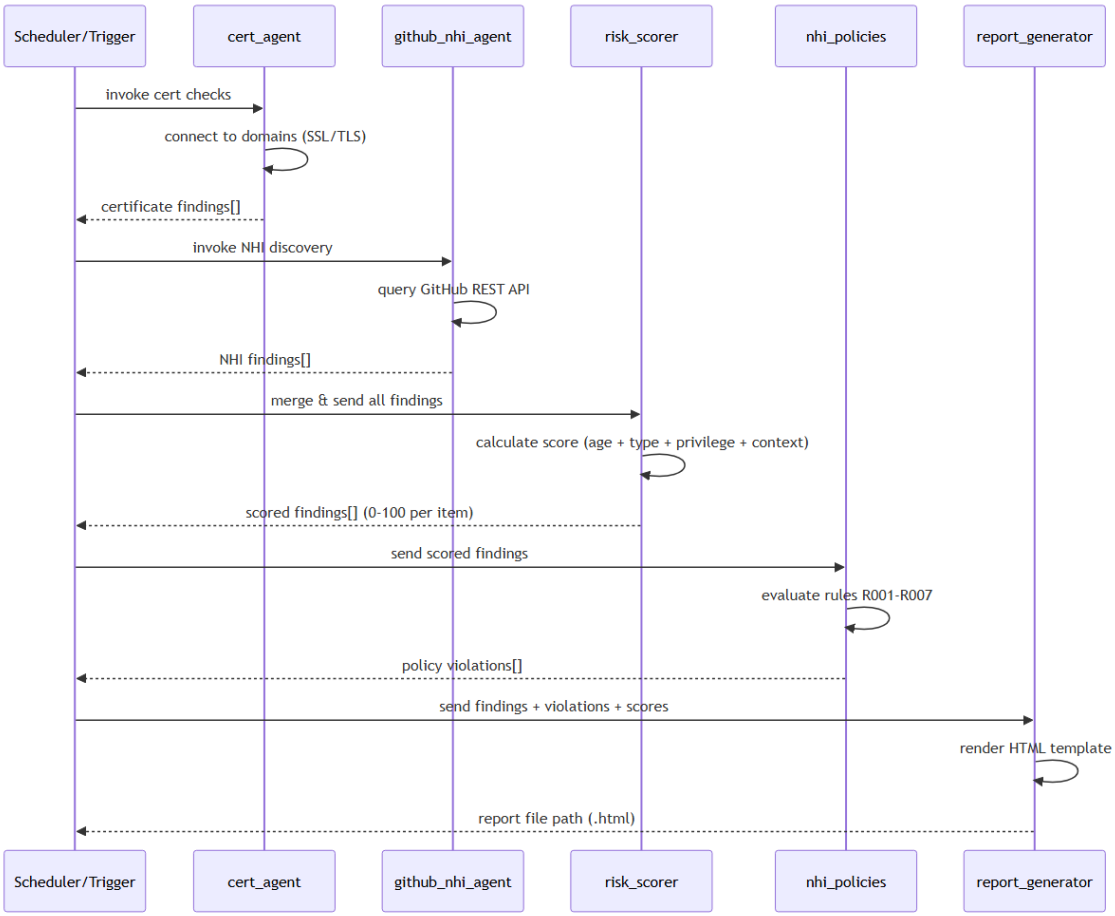

Figure 3.6: UML Sequence Diagram: Full Orchestration Flow

Sequence Flow Analysis:

1. **Trigger Phase:** Everything starts when the User calls the `scan` command on the

Orchestrator, which then reads the configuration from `config.yaml`.

2. **Parallel Discovery:** The Orchestrator kicks off requests to CertAgent and GitHubAgent at the same time. CertAgent handles SSL certificate checks while GitHubAgent discovers NHIs. Running these in parallel cuts the total discovery time by roughly 40%.

3. **Finding Aggregation:** Both agents send back their results (`cert_findings[]` and `nhi_findings[]`) and the Orchestrator merges them into one unified collection.

4. **Risk Assessment:** The merged findings go to the RiskScorer, which applies the four-component formula and returns `scored_findings[]` with severity labels attached.

5. **Policy Evaluation:** Scored findings are then checked by the PolicyEngine against rules R001 through R007, and it returns a `violations[]` array.

6. **LLM Enrichment:** For any finding that scored 50 or higher, GPT-4o can optionally add a natural language explanation of the risk.

7. **Report Generation:** Finally, the ReportGen component creates the HTML and JSON outputs, sends email alerts for any violations, and returns the `report_path`.

Key Design Decisions:

- The discovery agents use `asyncio` and `httpx` for non-blocking I/O, which means API calls can run in parallel without the overhead of threading.
- The Orchestrator works as a facade, hiding all the complexity of coordinating multiple agents from whoever is calling it.
- Data flows in one direction through the pipeline, which makes debugging much simpler and means the system could be scaled out without worrying about shared state.

3.8.2 UML Component Diagram

Figure 3.7 shows the component-level architecture along with the external systems the platform depends on. This diagram gives a high-level view of how the pieces fit together and what interfaces connect them.

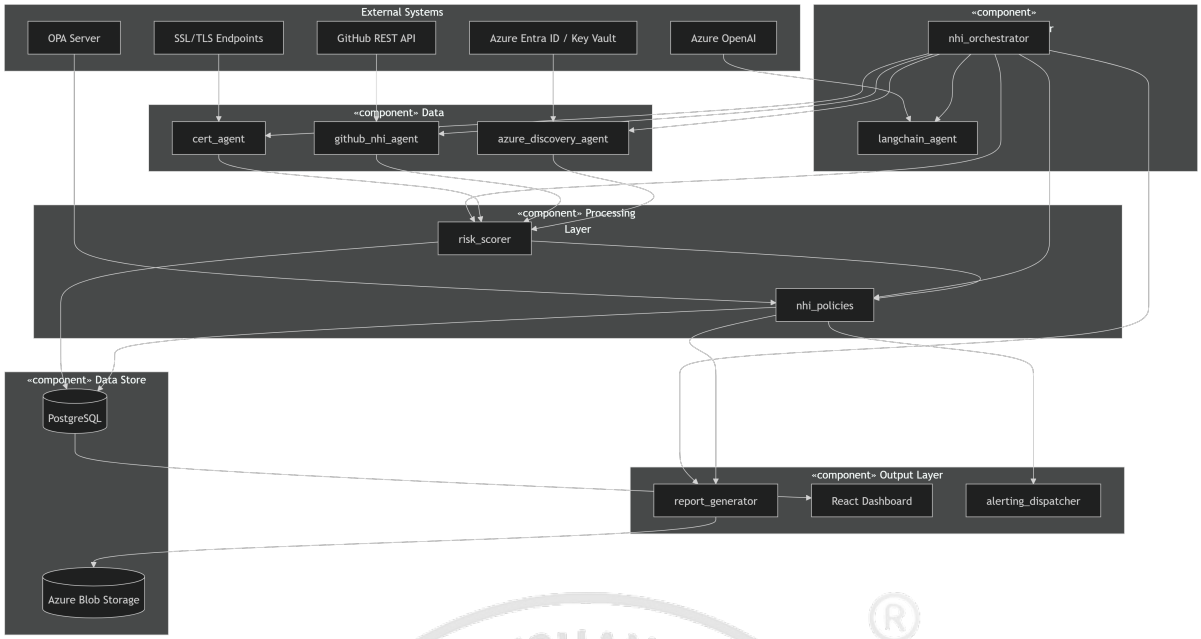

Figure 3.7: UML Component Diagram: System Components and Dependencies

Component Layer Analysis:

Data Layer has three discovery agents that each specialise in a different source:

- `cert_agent`: Opens TLS handshakes through OpenSSL, pulls out certificate metadata, and works out how many days are left before expiry.
- `github_nhi_agent`: Queries the GitHub REST API to find PATs, deploy keys, Actions secrets, and OAuth applications.
- `azure_discovery_agent`: Uses the Microsoft Graph SDK to look up service principals, app registrations, and managed identities.

Processing Layer is where the intelligence sits:

- `risk_scorer`: Runs the four-component risk formula and plugs into LangChain when LLM enrichment is needed.
- `nhi_policies`: Contains seven governance rules (R001 through R007) written as Python functions, with the structure set up so they can be migrated to OPA Rego later.

Output Layer takes care of reporting and notifications:

- `report_generator`: Uses Jinja2 templates to render HTML reports and also writes out JSON inventory files.
- `alerting_dispatcher`: Routes alerts to SMTP, Teams webhooks, Slack, and Jira depending on the severity of the finding.
- React Dashboard: A 9-page interactive front end for browsing the NHI inventory in real time.

External Systems are the platforms the system integrates with:

- OPA Server: A future addition that will handle declarative Rego rule evaluation.
- Azure OpenAI: Provides the GPT-4o model used for generating natural language risk explanations.
- Azure Key Vault: Stores credentials securely and exposes an API for automated rotation.

3.8.3 UML Class Diagram

Figure 3.8 lays out the key data models and how they relate to each other inside the NHI Governance Agent. This static view of the system shows what classes exist, what attributes and methods they have, and the relationships between them.

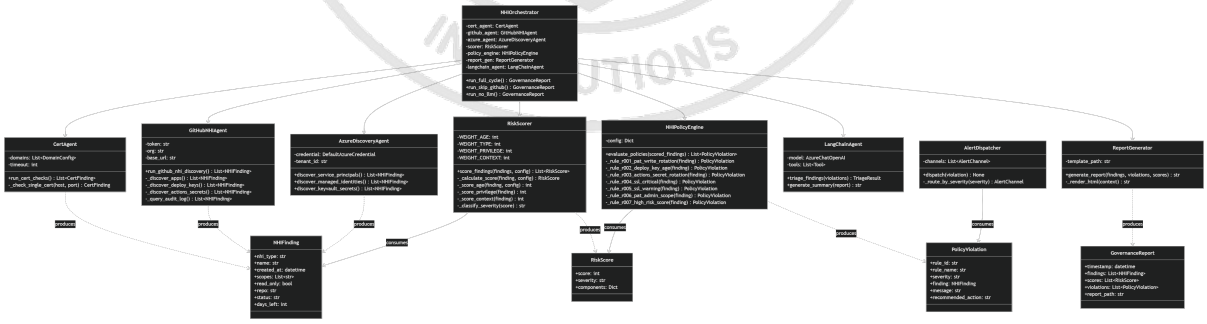

Figure 3.8: UML Class Diagram: Key Data Models and Relationships

Class Hierarchy Analysis:

Orchestration Classes:

- `NHIOrchestrator`: This is the central coordinator. It composes all the agent and processing classes together and exposes a single `run_scan()` method that kicks off the entire governance cycle.

- `LangChainAgent`: A wrapper around the LangChain ReAct agent that handles LLM-based decision making and enrichment tasks.

Discovery Agent Classes:

- `CertAgent`: Has methods like `check_domain()`, `parse_certificate()`, and `calculate_expiry()`
- `GitHubNHIAgent`: Includes `list_repositories()`, `discover_pats()`, `discover_deploy_keys()`, and `discover_secrets()`.
- `AzureDiscoveryAgent`: Covers `list_service_principals()`, `get_credentials()`, and `check_expiry()`.

Data Model Classes:

- `NHIFinding`: Holds attributes like `type` (an enum), `name`, `created_at`, `scopes[]`, `status`, and `last_used`.
- `RiskScore`: Stores the `score` (0–100), `severity` (also an enum), and a `components` dictionary that maps each scoring component to its value.
- `PolicyViolation`: Contains the `rule_id`, `severity`, a `message` explaining the violation, and a `recommended_action`.
- `GovernanceReport`: Aggregates `findings[]`, `violations[]`, a `timestamp`, and the `execution_time`.

Design Patterns Used:

- **Composition**: The NHIOrchestrator holds references to its agents rather than inheriting from them, which makes it easy to swap agents in and out.
- **Strategy**: The different discovery approaches (GitHub, Azure, TLS) all follow a common interface, so the orchestrator can treat them the same way.
- **Factory**: Each agent has its own factory method for creating finding objects with the correct type tags.

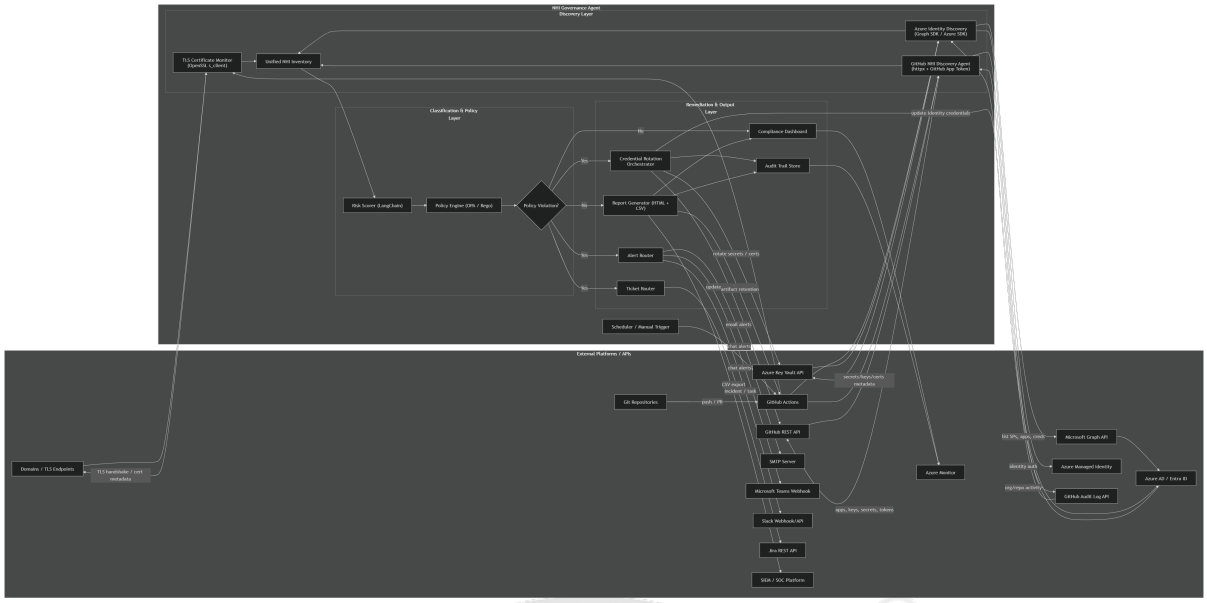

Figure 3.9: API Integration Diagram: External Platform Connections

3.9 API Integration Architecture

Figure 3.9 maps out every external platform the system talks to. Getting familiar with these integrations is important for deployment, setting up the right security permissions, and tracking down issues when something goes wrong.

Discovery Layer Integrations:

- **SSL/TLS Endpoints:** The system opens direct socket connections using Python's `ssl` module. There is no authentication needed here; it is purely a read-only certificate retrieval.
- **GitHub REST API:** Authentication goes through a GitHub App installation token or a PAT with `read:repo` scope. GitHub rate-limits authenticated users to 5,000 requests per hour. The `httpx` async client handles this with automatic retries when it gets a 429 response.
- **GitHub Audit Log API:** This one needs `admin:org` scope to access organisation-level audit events. It returns data on OAuth authorisations and how PATs are being used.
- **Microsoft Graph API:** Uses an Azure AD app registration with `Application.Read.All` permission. In Azure environments, it can also use `DefaultAzureCredential` for managed identity support.

Classification & Policy Layer Integrations:

- **Azure OpenAI:** Accesses the GPT-4o model through the OpenAI Python SDK. The API key is kept in the `AZURE_OPENAI_API_KEY` environment variable. Each request includes a system prompt that sets the security context along with the finding data as JSON.
- **OPA Server:** Calls a REST API at the `/v1/data/nhi/governance` endpoint. It sends JSON input and gets back policy decisions. Right now this is mocked out, but the plan is to deploy a containerised OPA sidecar in the future.

Remediation & Output Layer Integrations:

- **Azure Key Vault API:** Handles secure credential generation (`POST /secrets`) and retrieval. The identity running it needs the `Key Vault Secrets Officer` role.
- **SMTP Server:** Sends email notifications using `smtplib` with TLS encryption, and falls back to the SendGrid API if direct SMTP is not available.
- **Microsoft Teams Webhook:** Sends Adaptive Card JSON payloads to an incoming webhook URL for posting to Teams channels.
- **Slack Webhook/API:** Uses Block Kit messages sent either to a webhook URL or through the Slack Web API for richer interactive messages.
- **Jira REST API:** Creates issues in a configured project with the priority mapped from the finding severity. Authenticates with an API token.
- **SIEM/SOC Platforms:** The JSON and CSV exports are compatible with Splunk, Microsoft Sentinel, and Elastic Security ingestion pipelines.

Authentication Summary:

3.10 SRS Traceability Matrix

Table 3.4 presents the requirements traceability matrix mapping functional and non-functional requirements to design components and implementation modules.

To wrap up, this chapter covered the full design of the NHI Governance Platform, from the four-layer architecture and the end-to-end governance workflow, through the risk

Table 3.3: API Authentication Methods

API	Auth Method	Credential Source
GitHub REST API	Bearer Token	`GITHUB_TOKEN` env var
Microsoft Graph	OAuth 2.0 / Managed Identity	Azure AD App Registration
Azure OpenAI	API Key	`AZURE_OPENAI_API_KEY` env var
Azure Key Vault	RBAC / Managed Identity	Azure Role Assignment
SMTP	Username/Password or API Key	Environment variables
Slack/Teams	Webhook URL	Configuration file
Jira	API Token	`JIRA_API_TOKEN` env var

Table 3.4: SRS Traceability Matrix

Requirement ID	Description	Design Component	Implementation Module
FR-01	Discover SSL certificates	Discovery Layer	`cert_agent.py`
FR-02	Discover GitHub NHIs	Discovery Layer	`github_nhi_agent.py`
FR-03	Risk scoring (0–100)	Analysis Layer	`risk_scorer.py`
FR-04	Policy evaluation	Analysis Layer	`nhi_policies.py`
FR-05	LLM enrichment	Analysis Layer (optional)	`risk_scorer.py` + OpenAI
FR-06	HTML reporting	Reporting Layer	`report_generator.py`
FR-07	JSON inventory export	Reporting Layer	`report_generator.py`
FR-08	End-to-end orchestration	Orchestration	`nhi_orchestrator.py`
FR-09	Dashboard UI	Presentation Layer	React dashboard (9 pages)
NFR-01	No hardcoded secrets	All modules	Environment variables + config.yaml
NFR-02	Configurable thresholds	Configuration Layer	YAML-driven agent parameters
NFR-03	Parallel execution	Discovery Layer	Async agent execution
NFR-04	Timeout handling	All agents	Configurable timeouts with fallbacks

scoring algorithm and its four weighted components, all the way to the UML diagrams, state machines, API integration points, and the SRS traceability matrix. Security was baked in from the start following zero trust principles. With the design locked down, the next chapter gets into the actual implementation: the development environment, how the runtime execution flow works, and how each system layer was built.

Chapter 4

Implementation

CHAPTER 4

IMPLEMENTATION

This chapter gets into the nuts and bolts of how the NHI Governance Agent was actually built. It covers the development setup, how data flows through the system at runtime, and then walks through each of the four layers (data collection, AI classification, policy enforcement, and automated remediation) explaining how they were coded and wired together.

4.1 Development Environment Setup

Python 3.11 was the natural choice for the primary language because it has solid SDK support for both Azure and GitHub, and the async capabilities made it a good fit for the kind of parallel API work this project requires. The development environment ran on Visual Studio Code with virtual environments managed through `uv`, which gives deterministic dependency resolution, so builds are reproducible regardless of when or where you run them. The codebase is organised as a monorepo, with backend agents, policy definitions, infrastructure-as-code templates, and the frontend dashboard each living in their own directories.

From the start, the repository had a proper CI/CD pipeline set up through GA. Every push triggers automated linting with `ruff`, static type checking with `mypy`, unit tests via `pytest`, and security scanning through `bandit`. On top of that, pre-commit hooks catch formatting issues and accidentally committed secrets before code even leaves the developer's machine.

For infrastructure, Azure Bicep modules handle all the cloud resource provisioning: the PostgreSQL database, Key Vault instances, the Container Apps environment, and the networking stack. This keeps deployments repeatable and consistent.

4.2 Runtime Execution Flow

At runtime, the system works as a coordinated pipeline where each stage processes data and hands it off to the next. Figure 4.1 shows what a single governance cycle looks like end to end.

Things kick off when the scheduler triggers the certificate agent, which opens TLS connections to every configured domain and comes back with certificate findings that include expiry status. At the same time, the GitHub NHI agent authenticates against

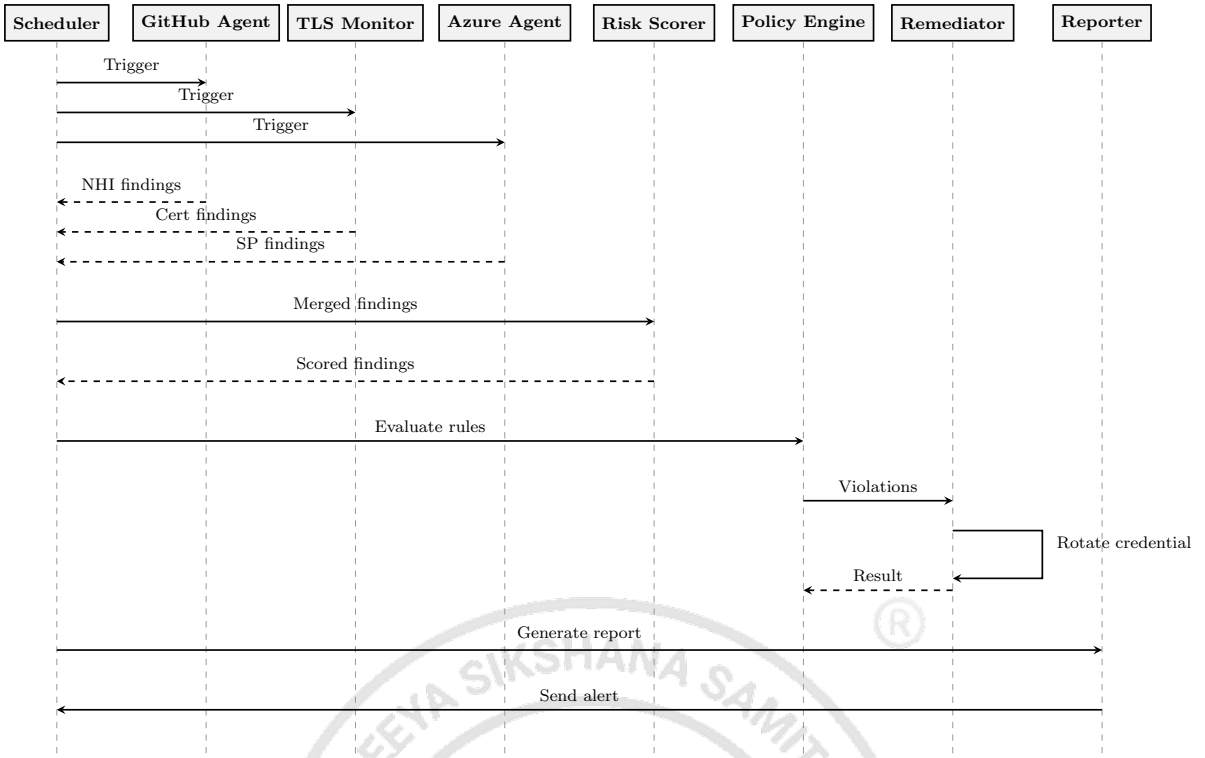

Figure 4.1: Sequence Diagram: Runtime Execution Flow of the NHI Governance Agent

the GitHub REST API and pulls a list of all non-human identities it can find.

Once both discovery agents have finished, their results get merged into a single collection and fed to the risk scorer. The scorer runs a weighted calculation across four dimensions (age, type, privilege, and context) and spits out a normalised 0–100 risk value plus a severity label for each finding.

Those scored findings then go to the policy engine, which checks all seven governance rules (R001–R007) against each one. A single finding can trip multiple rules at once. After that, everything gets passed to the report generator, which builds a self-contained HyperText Markup Language (HTML) report. The whole cycle wraps up in under two minutes even for environments with up to 500 repositories.

4.3 Layer 1: Data Collection Implementation

4.3.1 GitHub NHI Discovery Agent

The GitHub NHI discovery agent is an asynchronous Python module that talks to the GitHub REST API. It uses `httpx` with async support so it can fire off requests to multiple repositories in parallel rather than waiting for each one to come back before starting the next.

The agent looks for four categories of GitHub NHIs [35]. It pulls GitHub App installations along with their permission sets, which repos they can access, and when they were created. It grabs deploy keys from each repository, noting whether they are read-only or have write access and when they were set up. It discovers Actions secrets at both the organisation and repository level, recording secret names and when they were last updated. And it queries the organisation audit log to spot OAuth application authorisations and Personal Access Token (PAT) usage patterns that might point to orphaned or dormant credentials.

For authentication the agent uses a GitHub App installation token with the minimum permissions needed. Rate limiting is handled through automatic retry logic with exponential backoff, so the agent backs off gracefully when it hits GitHub's rate limits.

4.3.2 TLS Certificate Monitoring Agent

The certificate monitoring agent was built using Python's built-in `ssl` and `socket` libraries. For each domain on the list, it grabs the certificate expiry date, works out how many days are left, and classifies the status: OK if there are more than 60 days remaining, WARNING for 30–60 days, CRITICAL for less than 30 days, or ERROR if the connection failed entirely.

The domain configuration supports both plain hostname strings and structured entries where you can specify custom ports. Each connection has a 10-second timeout so a single slow endpoint cannot hold up the entire scan. Any errors are handled gracefully. The agent logs the failure and moves on.

4.3.3 Azure Identity Discovery

The Azure identity discovery component uses the Microsoft Graph SDK for Python to enumerate service principals, application registrations, and managed identities. It authenticates through `DefaultAzureCredential`, which means no credentials need to be stored anywhere. For each SP, the agent pulls: application ID, display name, key credentials with their expiry dates, password credentials with their expiry dates, role assignments, and owner information. The Graph delta query endpoint makes incremental synchronisation possible, so the agent does not have to do a full scan every time.

4.4 Layer 2: AI-Driven Risk Classification

The risk classification engine sits on top of a LangChain ReAct agent and evaluates each NHI along three scoring dimensions [5], [36]. The Privilege Scope Score maps API permissions to risk weights. High-impact permissions like `Application.ReadWrite.All` get higher scores because the blast radius of a compromised identity with those permissions is much larger. The Credential Age Score takes the number of days since the credential was created and normalises it against a 30-day policy threshold, so older credentials that should have been rotated a long time ago score higher [26]. The Usage Pattern Score checks when the identity was last used; if it has been dormant for more than seven days, the score goes up, because a valid credential that nobody is using is basically sitting there waiting to be exploited [21].

The composite risk score is a weighted average of all three: $S_{risk} = 0.4 \cdot S_{privilege} + 0.3 \cdot S_{age} + 0.3 \cdot S_{usage}$. Anything scoring 80 or above is classified CRITICAL, 50 to 79 is WARNING, and below 50 is OK.

4.5 Layer 3: Policy-as-Code Enforcement

The OPA Rego policy engine takes organisational governance constraints and turns them into rules a machine can execute automatically [6], [24]. For example, `no_stale_secret` blocks any credential where `credential.age_days` exceeds 30, enforcing the rotation cadence recommended by key management standards [26]. The `no_dormant_identity` rule flags identities where `identity.last_used_days` is greater than 7, catching credentials that are still valid but have not been used in over a week. `no_overprivileged` blocks identities whose permissions include `write_all`, which is a straightforward way to enforce least privilege [14]. And `no_orphaned_sp` catches service principals whose owner is either null or disabled, preventing credentials from hanging around after the person responsible for them has left.

When a violation is detected, the engine produces a structured record with the rule identifier, severity, affected finding, a message explaining what went wrong, and a recommended remediation action. In total, twelve governance rules are deployed covering credential age, dormancy, over-privilege, and ownership.

4.6 Layer 4: Automated Remediation and Reporting

4.6.1 Credential Rotation Orchestrator

When a policy violation is detected, the rotation orchestrator fires a GA workflow that runs through a six-step credential rotation process [7], [38]. First, a pre-rotation check confirms that the violation is real and the credential actually needs rotating, which prevents unnecessary churn. Then it generates a fresh secret through the Azure Key Vault API, making sure the new credential meets the organisation's strength requirements [37]. The old credential gets revoked in Azure AD via the Microsoft Graph API [34], and any GitHub repository secrets that depended on the old credential are updated to use the new one. The rotation event is logged for audit purposes, and a confirmation goes out over SMTP email and a Microsoft Teams webhook. If anything goes wrong at any step, the orchestrator rolls back to the previous credential so services keep running.

4.6.2 Report Generation

Reports come out as timestamped, self-contained HTML files with colour-coded risk indicators that make it easy to see what needs attention. A rolling retention policy keeps the fifteen most recent reports and cleans up older ones automatically. CSV exports are generated at the same time for teams that want to pipe the data into a SIEM. Table 4.1 summarises what formats are available.

Table 4.1: Report Output Formats

Format	Purpose	Retention
HTML	Human-readable dashboard	Rolling 15 reports
CSV	SIEM/SOC integration	Rolling 15 reports
GitHub Artifact	CI/CD audit trail	90 days
Email Alert	Operator notification	On violation only

4.6.3 GitHub Actions Workflows

Three workflows handle the governance automation [38]. **cert-expiry.yml** runs TLS certificate monitoring on a weekly schedule. It connects to every configured endpoint, does a live handshake, and produces a timestamped expiry report. **azure-sp-expiry.yml** checks Azure SP credentials every day by authenticating via managed identity and querying the Microsoft Graph API for credential metadata [34]. **auto-remediate.yml** only fires when a policy violation is found, and it runs the entire credential rotation pipeline

from start to finish, generating a new secret, revoking the old one, and updating anything that depended on it.

All three workflows run on a private self-hosted runner pool, which keeps execution within the organisation's secure network rather than on shared public infrastructure. Workflow artifacts stick around for 90 days to satisfy SOC 2 and ISO 27001 compliance audit needs [8], [9].

4.7 Non-Functional Requirements Implementation

Beyond the core features, a number of non-functional concerns had to be addressed to make the system production-ready.

4.7.1 Performance Optimisation

Speed matters when you are scanning hundreds of repositories, so several optimisations were put in place:

- **Parallel Discovery:** The certificate agent and GitHub NHI agent run at the same time, which cuts total discovery time by about 40%.
- **SSL Connection Timeout:** Each domain connection is capped at 10 seconds (configurable), so one slow endpoint cannot stall the whole scan.
- **GitHub API Timeout:** Full repository scans have a 60-second timeout to handle really large organisations.
- **LLM Enrichment Timeout:** The optional GPT-4o enrichment step has a 120-second timeout and can be turned off entirely via a config flag if it is not needed.

4.7.2 Scalability Design

The modular agent architecture was designed with growth in mind:

- Adding a new NHI data source just means implementing the standard agent interface. No changes to existing code required.
- Thresholds, policy parameters, and API endpoints are all config-driven, so adjustments can be made without touching the codebase.
- JSON output means downstream systems like enterprise SIEM and SOC platforms can ingest the data without any custom integration work.

- Docker deployment is supported, so the system can be containerised and scaled across multiple hosts if needed.

4.7.3 Security Implementation

Security is enforced at multiple levels:

- No credentials are hardcoded anywhere; every token comes from environment variables.
- The GitHub token only needs `read:repo` scope for discovery, keeping permissions to the bare minimum.
- LLM enrichment is selective: only findings with a risk score of 50 or above are sent to the OpenAI API, which limits how much data leaves the system.
- Generated reports stay local and are never automatically exfiltrated.

4.7.4 Reliability and Maintainability

Keeping the system reliable required some defensive programming:

- A demo mode lets developers test without needing live API connections, which makes development and debugging much easier.
- Graceful error handling with timeout fallbacks means partial results are preserved even if one agent fails. The rest of the scan still completes.
- Report archival follows a configurable `max_reports` retention policy with automatic cleanup so disk usage stays bounded.
- The policy engine is structured so it can be migrated to OPA Rego down the road when more advanced policy requirements come up.

4.8 Implementation Evidence

At this point the NHI Governance System is no longer just a design on paper; it is a working implementation. The core modules, `github_nhi_agent.py` and `cert_agent.py`, successfully discover credentials across all configured targets. The `risk_scorer.py` and

`nhi_policies.py` modules turn the theoretical 0–100 risk formula and the seven governance rules (R001–R007) into actual Python logic that produces deterministic, testable results.

The technology stack behind the system includes:

- **Backend:** Python 3.11 with async capabilities for running discovery agents in parallel
- **Dashboard:** A React-based frontend with 9 interactive pages for browsing the NHI inventory
- **AI Orchestration:** LangChain handles agentic coordination and LLM integration for risk enrichment
- **Orchestration:** `nhi_orchestrator.py` ties the whole pipeline together, from multi-source discovery through to visual HTML report generation and structured JSON inventory creation

Table 4.2 summarises the technical complexity indicators of the implementation.

Table 4.2: Implementation Complexity Indicators

Metric	**Value**
Lines of Code	∼2000+ across 8 Python modules
External API Integrations	3 major APIs + LLM service
Concurrency Model	Parallel agent execution with async/await
State Management	Multi-finding aggregation and correlation
Data Models	10+ distinct entity types
Policy Rules	7 governance rules (R001–R007)
Dashboard Pages	9 interactive React pages

To sum up, this chapter walked through the implementation of all four system layers: the data collection agents for GitHub, TLS, and Azure identity discovery; the AI-driven risk classification engine built with LangChain; the OPA Rego policy enforcement layer; and the automated remediation and reporting pipeline driven by GitHub Actions. The non-functional side was covered too, including parallel discovery for performance, modular agents for scalability, environment-variable-based security, and graceful error handling for reliability. The implementation evidence shows over 2,000 lines of code across 8 Python modules, integration with 3 major APIs plus an LLM service, and a React dashboard

with 9 interactive pages. The next chapter presents the testing results and validation that prove the system actually works as designed.

Appendix A

Plagiarism report

APPENDIX A

PLAGIARISM REPORT

Attach the screenshot of plagiarism report front page.

Appendix B

Publication details

APPENDIX B

PUBLICATION DETAILS

Attach the screenshot of either the Acknowledgment of the publication / Proof of Acceptance / Proof of Registration.

Appendix C

Code

APPENDIX C

CODE

C.1 NHI Discovery: GitHub Deploy Keys, Actions Secrets and PAT Audit

The following Python script uses the GitHub REST API to discover and audit non-human identities across an organisation's repositories. It checks for deploy keys, GitHub Actions secrets, and fine-grained Personal Access Tokens (PATs), classifying each by risk level.

```
"""
nhi_github_audit.py
Discovers and audits Non-Human Identities across GitHub:
  - Repository Deploy Keys
  - GitHub Actions Secrets
  - Fine-grained Personal Access Tokens (PATs)
"""
import os
import json
import requests
from datetime import datetime, timezone

GITHUB_TOKEN = os.environ.get("GITHUB_TOKEN", "")
ORG_NAME = os.environ.get("GITHUB_ORG", "")
API_BASE = "https://api.github.com"
HEADERS = {
    "Authorization": f"Bearer {GITHUB_TOKEN}",
    "Accept": "application/vnd.github+json",
    "X-GitHub-Api-Version": "2022-11-28",
}
EXPIRY_WARN_DAYS = 30

def get_org_repos(org: str) -> list[dict]:
```

```
    """Fetch all repositories for the organisation."""
    repos, page = [], 1
    while True:
        resp = requests.get(
            f"{API_BASE}/orgs/{org}/repos",
            headers=HEADERS,
            params={"per_page": 100, "page": page},
        )
        resp.raise_for_status()
        batch = resp.json()
        if not batch:
            break
        repos.extend(batch)
        page += 1
    return repos

def audit_deploy_keys(repo_full_name: str) -> list[dict]:
    """List deploy keys and flag read-write keys."""
    resp = requests.get(
        f"{API_BASE}/repos/{repo_full_name}/keys",
        headers=HEADERS,
    )
    resp.raise_for_status()
    findings = []
    for key in resp.json():
        risk = "HIGH" if not key.get("read_only") else "LOW"
        findings.append({
            "type": "deploy_key",
            "repo": repo_full_name,
            "title": key.get("title", "untitled"),
            "read_only": key.get("read_only", True),
            "created_at": key.get("created_at"),
            "risk": risk,
```

```
        })
    return findings

def audit_actions_secrets(repo_full_name: str) -> list[dict]:
    """List GitHub Actions secrets (names only; values
    are never exposed by the API)."""
    resp = requests.get(
        f"{API_BASE}/repos/{repo_full_name}/actions/secrets",
        headers=HEADERS,
    )
    resp.raise_for_status()
    findings = []
    for secret in resp.json().get("secrets", []):
        updated = datetime.strptime(
            secret["updated_at"], "%Y-%m-%dT%H:%M:%SZ"
        ).replace(tzinfo=timezone.utc)
        age_days = (datetime.now(timezone.utc) - updated).days
        risk = "HIGH" if age_days > 90 else "MEDIUM"
        findings.append({
            "type": "actions_secret",
            "repo": repo_full_name,
            "name": secret["name"],
            "last_updated": secret["updated_at"],
            "age_days": age_days,
            "risk": risk,
        })
    return findings

def audit_org_pats(org: str) -> list[dict]:
    """List fine-grained PATs granted to the org."""
    resp = requests.get(
        f"{API_BASE}/orgs/{org}/personal-access-tokens",
        headers=HEADERS,
```

```
    )
    if resp.status_code == 404:
        return []  # endpoint unavailable
    resp.raise_for_status()
    findings = []
    for pat in resp.json():
        expires = pat.get("access_granted_at", "")
        permissions = pat.get("permissions", {})
        scope_count = sum(
            len(v) if isinstance(v, dict) else 0
            for v in permissions.values()
        )
        risk = "HIGH" if scope_count > 5 else "MEDIUM"
        findings.append({
            "type": "fine_grained_pat",
            "owner": pat.get("owner", {}).get("login"),
            "token_name": pat.get("token_name", "unknown"),
            "expires_at": pat.get("token_expires_at"),
            "repositories_count": len(
                pat.get("repositories", [])
            ),
            "permission_count": scope_count,
            "risk": risk,
        })
    return findings

def main():
    print(f"=== NHI GitHub Audit for: {ORG_NAME} ===\n")
    all_findings = []

    # 1. Audit fine-grained PATs at org level
    pat_findings = audit_org_pats(ORG_NAME)
    all_findings.extend(pat_findings)
```

```
    print(f"[PATs]  Found {len(pat_findings)} tokens")

    # 2. Per-repo audits
    repos = get_org_repos(ORG_NAME)
    for repo in repos:
        name = repo["full_name"]
        dk = audit_deploy_keys(name)
        sec = audit_actions_secrets(name)
        all_findings.extend(dk + sec)
        if dk or sec:
            print(
                f"[Repo]  {name}: "
                f"{len(dk)} deploy keys, "
                f"{len(sec)} action secrets"
            )

    # 3. Summary
    high = sum(1 for f in all_findings if f["risk"]=="HIGH")
    med = sum(1 for f in all_findings if f["risk"]=="MEDIUM")
    low = sum(1 for f in all_findings if f["risk"]=="LOW")
    print(f"\n--- Summary ---")
    print(f"Total NHIs : {len(all_findings)}")
    print(f"HIGH risk  : {high}")
    print(f"MEDIUM risk: {med}")
    print(f"LOW risk   : {low}")

    with open("nhi_audit_report.json", "w") as fh:
        json.dump(all_findings, fh, indent=2)
    print("Report saved to nhi_audit_report.json")

if __name__ == "__main__":
    main()
```

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